

SC2014 : Sensors, Interfacing & Digital Control

Part 2 : Tutorial 3 – Intro to Digital Control System

Q1 : Use Initial Value Theorem to calculate the value of $x(0)$ for the following system transfer functions (Z-Transform) :

i)
$$X(z) = \frac{z^2 + z + 1}{(z+2)(z+1)}$$

Theorem: $x(0) = \lim_{n \rightarrow 0} x(n) = \lim_{z \rightarrow \infty} X(z)$

push it into $1/z$ as $1/\infty$ is 0.

eg i) = $(z^2(1+1/z+1/z^2))/(z(1+2/z)z(1+1/z))$, if $z \rightarrow \infty$ then $z^2/z^2 = 1$

ii)
$$X(z) = \frac{z+4}{(z+2)(z+1)}$$

Order of magnitude, ii) == 0

Q2 : Use Final Value Theorem to calculate the value of $x(\infty)$ for the following system transfer functions (Z-Transform) :

Theorem: $x(\infty) = \lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} ((z-1)X(z))$

i)
$$X(z) = \frac{z^2}{(z-1)(z-0.3)}$$

i) $(z-1)X(z) = z^2/(z-0.3)$, let $z = 1$, then $x(\infty) = 1.43$

ii)
$$X(z) = \frac{z+1}{3(z-1)(z+0.4)}$$

ii) 0.48

Q3 : Consider a system described by the following difference equations :
where $y(k)$, $x(k)$ are the output/input of the system.

$$y(k) + 2.1y(k-1) + 1.4y(k-2) + 0.32y(k-3) = x(k)$$

Difference eqn: in terms of $y(k)$ and $x(k)$

Differential eqn: in terms of dy/dt etc.

Z transform: in terms of z^{-1} etc.

If the input to this system is an impulse function, find the resultant output, $Y(z)$.

Q4 : A mechanical lifting system (with the following transfer function) is shown below :

$$T.F. = \frac{Y(z)}{X(z)} = \frac{1}{z(z+8)(z+1)}$$

If the input to this system is a sampled step function, find the difference equation that you will need to implement for a software simulation of this system.